

Alberto Reyes Castro

Physics Graduate & MSc Artificial Intelligence Technology Student



About me

Physics graduate and MSc Artificial Intelligence Technology student with experience in Python automation, machine learning, IoT systems, scientific communication, LaTeX and technical documentation. Interested in applying AI, data-driven methods and scientific computing to real-world and interdisciplinary problems.

Specializations

• Machine Learning • Data Analysis • Scientific Computing • AIoT and TinyML • Technical Documentation • Scientific Communication • Python Programming • LaTeX

Skills

Python Automation / Machine Learning / Data Analysis / Scientific Computing / IoT Systems / GPU-Accelerated Whisper Transcription / Subtitle Generation / Regex / Fuzzy Matching / Technical Documentation / Scientific Writing

Languages

Spanish – Native
English – Advanced C1
Japanese – JLPT N3 level
French – Limited working proficiency

SUMMARY

Physics graduate with strong foundations in mathematics, programming, scientific communication and core areas of physics. Currently pursuing a two-year, full-time MSc in Artificial Intelligence Technology with Advanced Practice at Northumbria University, London Campus, with interests in machine learning, AIoT, scientific computing, data analysis and applied project-based work.

Achieved Distinction-level marks of 70+/100 across all completed first-semester MSc final assessments and projects, with assessed work spanning machine learning, AI project development, leadership and professional practice in computing.

Completed a Final Degree Project in English on the structural mechanisms of ferroelectricity and ferroelasticity in β' -RE₂(MoO₄)₃ compounds, achieving a high distinction. The project involved crystallographic analysis using neutron and X-ray diffraction data and has attracted interest for potential development into a scientific publication.

Received the Academic Merit Convocation award from Universidad de La Laguna, an annual distinction granted to students with outstanding academic performance and ranking among the top of their graduating class.

EXPERIENCE

Mar 2025–May 2025

Student Internship – Department of Communication and Scientific Culture (UC3)

INSTITUTO DE ASTROFÍSICA DE CANARIAS (IAC) · La Laguna 

Contributed to bilingual science communication and outreach at the IAC. Developed a Python-based, AI-assisted transcription workflow using GPU-accelerated Whisper to generate segmented multilingual subtitles. Built a semi-automated text-matching pipeline using regex and fuzzy matching to link outreach articles with scientific publications. Supported public outreach events, audiovisual production, citizen surveys, interviews with Nobel Laureates in Physics, image analysis with Peter-Soft, and technical reporting in LaTeX.



Jan 2012–Jan 2016

Basketball Player

REAL CLUB NAÚTICO DE TENERIFE (RCNT) · Santa Cruz de Tenerife 

Competed in regional leagues and national tournaments, developing teamwork, discipline, leadership and performance under pressure.



EDUCATION

Jan 2026–Jan 2028

MSc Artificial Intelligence Technology with Advanced Practice

NORTHUMBRIA UNIVERSITY, LONDON CAMPUS · London, UK 

Distinction-level marks of 70+/100 across all completed first-semester final assessments and projects.



Sep 2018–Jul 2025

BSc in Physics

UNIVERSIDAD DE LA LAGUNA · Tenerife, Spain 

Academic Merit Convocation award. Final Degree Project completed in English with high distinction.



Sep 2022–Oct 2022

Intensive Japanese Course (4 weeks)

EDUCATION FIRST LANGUAGE ABROAD · Tokyo, Japan 



Jul 2017

Intensive English Course (4 weeks)

EDUCATION FIRST LANGUAGE ABROAD · Brighton, UK 



SELECTED TECHNICAL PROJECTS

- 2026** **AI-Enhanced Smart Home System** – Modular AIoT prototype using ESP32-S3, Wokwi, Blynk, Edge Impulse, rule-based automation and lightweight environmental classification.
- 2026** **Credit Card Fraud Detection using Machine Learning** – MSc machine-learning report focused on imbalanced classification, preprocessing, resampling strategies, model comparison, threshold selection and evaluation metrics. Distinction mark: 85/100.
- 2026** **Computational Physics Numerical Methods** – Scientific Python project implementing interpolation, numerical integration, ODE solvers and stochastic simulation with NumPy, SciPy and Matplotlib.
- 2025** **IAC Publication–Press Note Matching** – AI-assisted workflow for matching IAC scientific publications with press notes using regex, fuzzy matching and reproducible notebook-based analysis.
- 2025** **Adaptive Whisper Transcription Pipeline** – Portable CPU/GPU transcription workflow for audio and video files using faster-whisper, automatic hardware selection, safe input handling and optional SRT subtitle generation.

CERTIFICATES AND ADDITIONAL TRAINING

- 2025** AI & Big Data in Biomedical and Healthcare Applications (300h) – DYM Canarias
- 2025** IELTS Academic UKVI (7.0/9.0, C1) – British Council
- 2025** NASA Open Science 101 – NASA
- 2024** Introduction to Data Science – Santander
- 2024** Artificial Intelligence and Productivity – Santander
- 2023** IDEAUP Open Innovation Training – Wrocław University of Science and Technology
- 2022** Cambridge Advanced (C1) – University of Cambridge
- 2022** Basic CPR and First Aid – Cabildo de Tenerife
- 2019** COEFIS Physics Student Congress – Universidad de La Laguna
- 2017** Scientific Week Certificate – CSIC

ACADEMIC PROJECTS AND OUTREACH CONTRIBUTIONS

- 2025** *Final Degree Project: Structural Mechanisms of Ferroelectricity and Ferroelasticity in Compounds with the Formula: β^I -RE₂(MoO₄)₃*. Manuscript potential based on this work.
- 2025** *El eclipse parcial que recordó a Canarias el valor de mirar al cielo*, **Astronomía Magazine**
- 2025** *The WEAVE spectrograph captures the interaction of supernova SN 2023ixf with its surroundings one year after its explosion*
IAC News
- 2025** *Partial Solar Eclipse of March 2025*, **IAC News**

CONTACT DETAILS & LINKS

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